

Higher Apps September Progress Test - Mixed Revision Questions

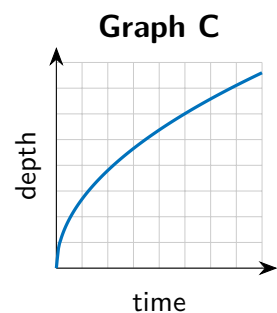
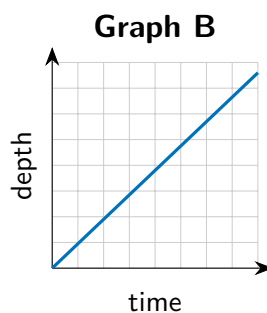
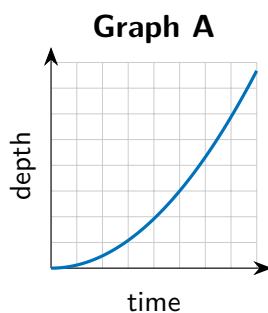
1. An employee earns an annual salary of £37,800. They also receive commission equal to 6% of their monthly sales and a bonus of £150 in any month in which their sales exceed £8,000.

In June, the employee makes sales worth £9,400.

Calculate the employee's gross pay for June.

2. Water is poured into an empty container at a constant rate. The container is narrow at the bottom and gradually becomes wider towards the top.

The graphs below show three possible relationships between the time since pouring began and the depth of the water.



State which of Graph A, Graph B or Graph C provides the most plausible model for the relationship. Explain your answer.

3. A survey of 200 students recorded whether they study Mathematics or Computing. The results are shown in the table.

	Mathematics	Not Mathematics
Computing	54	36
Not Computing	62	48

One student is selected at random. Determine:

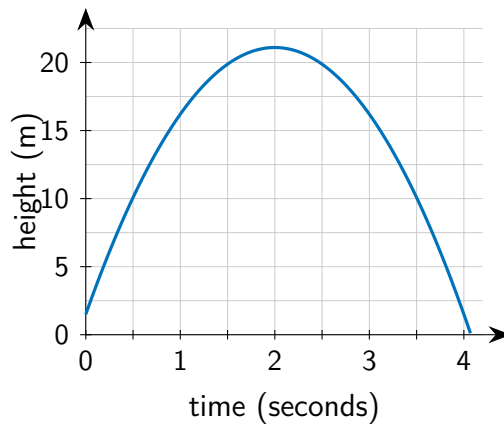
- (a) $P(\text{Mathematics})$;
 - (b) $P(\text{Mathematics and Computing})$;
 - (c) $P(\text{Mathematics or Computing})$;
4. Estimate the total number of hours a typical secondary school teacher spends marking work during a their career.

State clearly any assumptions you make.

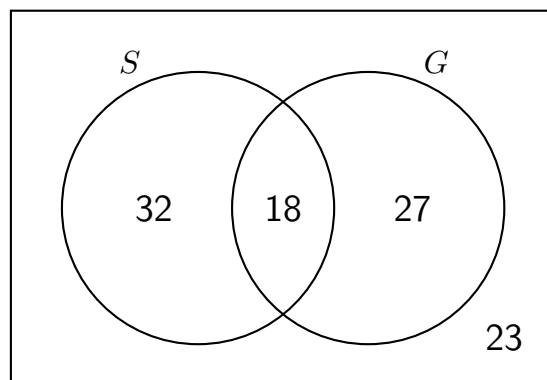
5. The height, h metres, of a ball t seconds after it is thrown is modelled by

$$h = -4.9t^2 + 19.6t + 1.5.$$

The graph of the model is shown below.



- (a) State the type of relationship shown.
 - (b) State the dependent variable.
 - (c) Calculate the height of the ball after 3 seconds.
 - (d) Calculate the greatest height reached by the ball.
6. A group of 100 young people were asked whether they use a streaming service, S , or a gaming service, G . The completed Venn diagram shows the results.



One person is selected at random. Determine:

- (a) $P(G)$;
- (b) $P(S \text{ and } G)$;
- (c) $P(S \text{ or } G)$;
- (d) the probability that the person uses the streaming service but not the gaming service;
- (e) the probability that the person uses exactly one of the services;

Data Booklet: Scottish tax bands

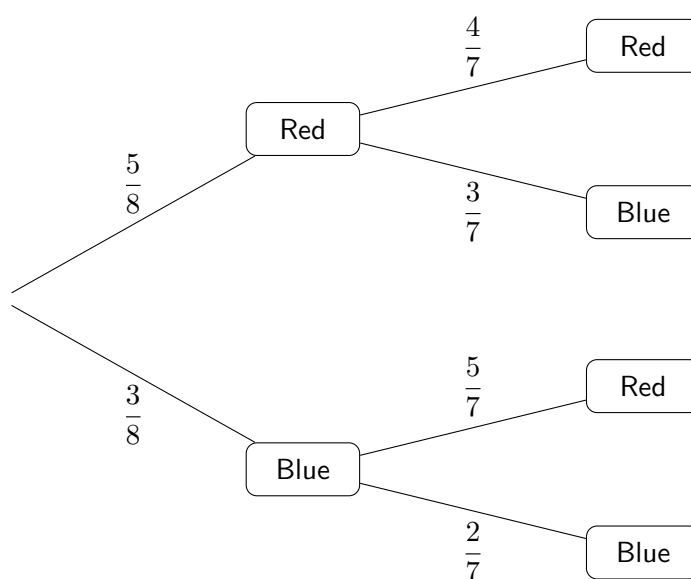
Band	Taxable Income	Scottish tax rate
Personal Allowance	Up to £12,570	0%
Starter rate	£12,570 – £15,397	19%
Basic rate	£15,397 – £27,491	20%
Intermediate rate	£27,491 – £43,662	21%
Higher rate	£43,662 – £75,000	42%
Advanced rate	£75,000 – £125,140	45%
Top rate	Over £125,140	48%

7. Erin earns an annual salary of £48,600. She contributes 4.5% of her gross salary to her workplace pension. Her employer contributes a further 7% of her gross salary.

- Calculate the total amount paid into Erin's pension fund each year.
- Using the extract from the Data Booklet above, calculate the amount of income tax Erin will pay each month.

8. A bag contains 5 red counters and 3 blue counters. Two counters are selected at random without replacement.

The tree diagram represents the possible outcomes.

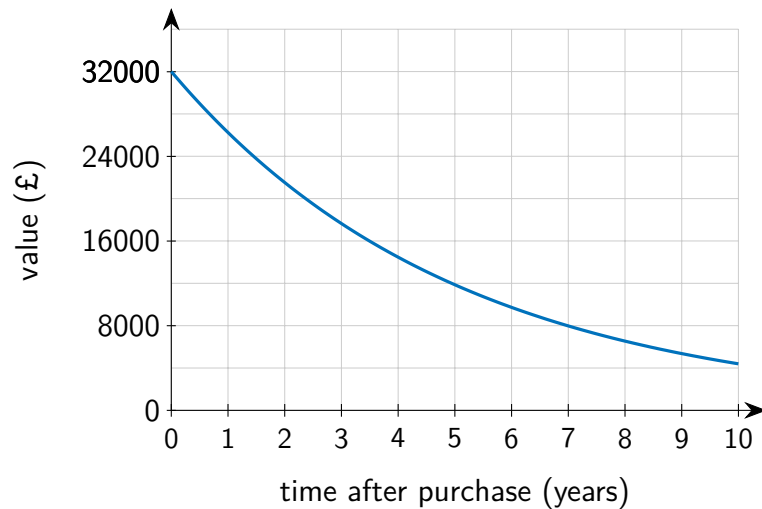


- Explain why the probability of selecting a red counter on the second selection is $\frac{4}{7}$ when the first counter selected is red.
- Calculate the probability that both counters selected are red.
- Calculate the probability that the two counters are different colours.
- Calculate the probability that at least one blue counter is selected.

9. The value, V pounds, of a machine t years after it is purchased is modelled by

$$V = 32000 \times 0.82^t.$$

The graph of the model is shown below.



- (a) State whether the model represents exponential growth or exponential decay.
 - (b) State the independent variable.
 - (c) State the initial value of the machine.
 - (d) Calculate the value of the machine after 4 years 6 months.
10. An employee is paid £17.20 per hour for the first 37 hours worked each week. Any additional hours are paid at time-and-a-half.
- In one week, the employee works 45 hours and receives commission equal to 4% of sales worth £2,750. Calculate the employee's total gross pay for the week.
- (a) Calculate the employee's total gross pay for the week.
 - (b) Stating any assumptions required, estimate the gross pay the employee expect to earn in a year.
11. The probability that a pupil travels to school by bus is 0.55.
- If the pupil travels by bus, the probability that they arrive late is 0.12. If the pupil does not travel by bus, the probability that they arrive late is 0.04.
- (a) Draw a tree diagram to represent this information.
Label every branch with the appropriate probability.
 - (b) Calculate the probability that the pupil travels by bus and arrives late.
 - (c) Calculate the probability that the pupil arrives late.

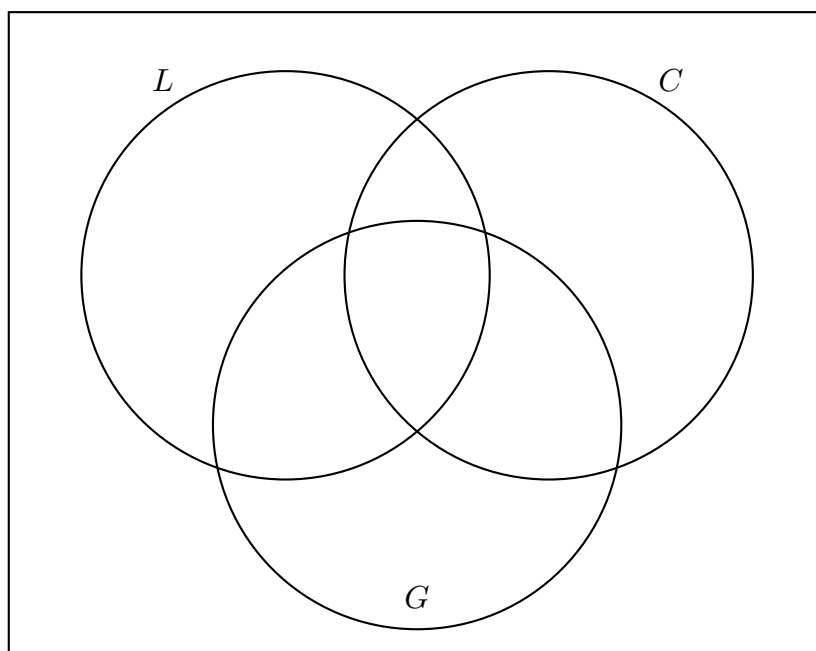
12. A group of 150 college students were asked which of the following facilities they use:

- the library, L ;
- the café, C ;
- the gym, G .

The following information was collected:

- 78 students use the library.
- 70 students use the café.
- 55 students use the gym.
- 32 students use both the library and the café.
- 27 students use both the library and the gym.
- 21 students use both the café and the gym.
- 8 students use all three facilities.

The numbers given for students using two facilities include students who use all three facilities.

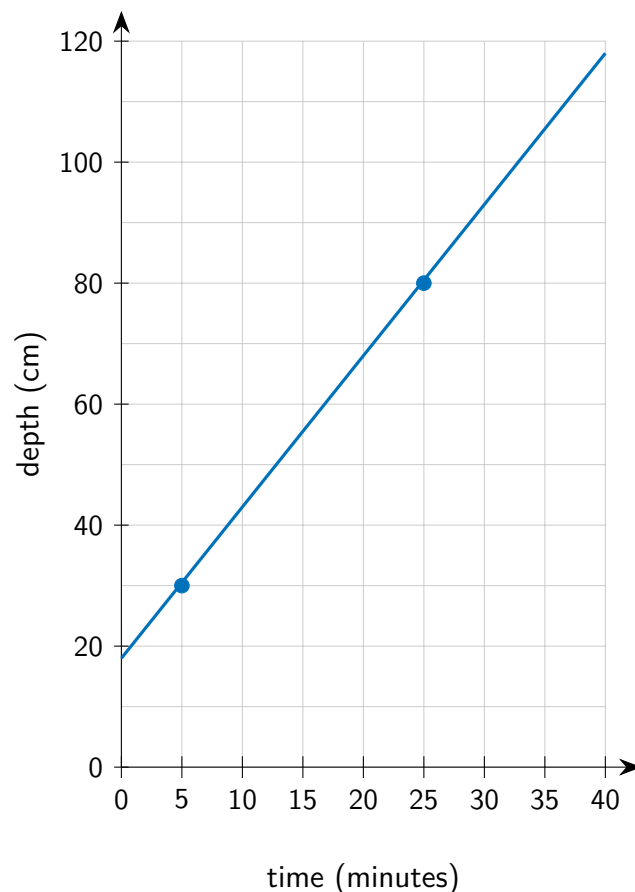


(a) Complete the Venn diagram showing the number of students in each region.

(b) One student is selected at random. Determine the probability that the student:

- uses only the library;
- uses exactly two facilities;
- does not use the café;
- uses at least one facility.

13. The graph shows the depth, d centimetres, of water in a tank t minutes after a tap is opened.



- (a) State the type of relationship shown.
- (b) State the independent variable.
- (c) State the dependent variable.
- (d) Calculate the rate of change, including appropriate units.
- (e) Use the graph to estimate the depth of the water after half an hour.

14. The average cost, A pounds, of manufacturing an item is modelled by

$$A = 3x^2 - 36x + 155,$$

where x is the number of thousands of items manufactured.

- (a) State the type of relationship the model uses.
- (b) Calculate the average cost when 4,000 items are manufactured.

15. Estimate the total volume of water used for showers in a town of 5,000 people during one week.
Give your answer in litres and state clearly any assumptions you make.

Data Booklet: National Insurance contributions

You begin paying National Insurance once you earn more than £1048 a month (this is the amount for the 2025/26 tax year).

For payslips dated between 6 April 2025 and 5 April 2026, you pay 8% of your monthly earnings between £1048 and £4189; 2% of your monthly earnings above £4189.

National Insurance is calculated on a person's salary before deductions such as pension contributions.

16. Sal earns a monthly gross salary of £5,134 and pays a percentage of this into his pension fund.
- (a) Explain why Sal's monthly National Insurance contribution can be calculated without knowing how much he pays into his pension fund.
 - (b) Calculate Sal's monthly National Insurance contribution.

17. The number of subscribers S to an online service is modelled by

$$S = 850 \times 1.12^m,$$

where m is the number of months after the service is launched.

- (a) State the independent variable.
 - (b) State the dependent variable.
 - (c) State what kind of model is being used to describe the relationship.
 - (d) Find the number of subscribers when the service is initially launched.
 - (e) Calculate the predicted number of subscribers after 9 months.
18. Ross earns an annual salary of £54,000. He contributes £3,600 of his gross annual salary to a workplace pension, and pays £257.55 each **month** in National Insurance contributions.
- (a) Calculate Ross' annual taxable pay.
 - (b) Calculate Ross' annual income tax.
 - (c) Calculate Ross' annual net pay.

Use the annual Scottish income tax bands given before question 7.

19. The weekly profit, P pounds, made by a food stall is modelled by

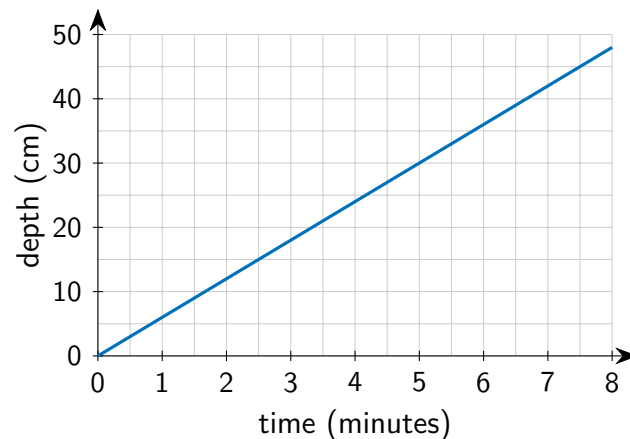
$$P = -25(w - 6)^2 + 950,$$

where w is the number of weeks after the stall opens.

- (a) State the independent variable.
- (b) State the dependent variable.
- (c) Calculate the profit after 2 weeks.

20. A cylindrical tank is initially empty. Water is pumped into the tank at a constant rate.

The graph models the relationship between the time since pumping began and the depth of water in the tank.



- State the type of relationship shown.
- Calculate the rate of change, giving suitable units in your answer.
- Explain how the graph would change if water were pumped into the tank at a faster constant rate.
- Explain how the graph would change if there were already 10 centimetres of water in the tank when pumping began.

ANSWERS

- £3,864
- Graph C. The water depth increases quickly at first because the bottom is narrow. As the container becomes wider, a larger volume of water is required for the same increase in depth, so the rate at which the depth increases gradually decreases.
- $\frac{116}{200} = \frac{29}{50}$
 - $\frac{54}{200} = \frac{27}{100}$
 - $\frac{152}{200} = \frac{19}{25}$
- For example, assume that a teacher:
 - spends approximately 5 hours marking each week;
 - marks for 39 weeks each year;
 - has a 35-year teaching career.

$$5 \times 39 \times 35 = 6825$$

Approximately 6800 hours.

5. (a) Quadratic
 (b) Height, h , measured in metres
 (c) 16.2 metres
 (d) 21.1 metres
6. (a) $\frac{45}{100} = \frac{9}{20}$
 (b) $\frac{18}{100} = \frac{9}{50}$
 (c) $\frac{77}{100}$
 (d) $\frac{32}{100} = \frac{8}{25}$
 (e) $\frac{59}{100}$
7. (a) £5,589
 (b) £625.61 per month
8. (a) After one red counter has been selected, there are 4 red counters remaining from a total of 7 counters. Therefore, the probability is $\frac{4}{7}$.
 (b) $\frac{5}{14}$
 (c) $\frac{15}{28}$
 (d) $\frac{9}{14}$
9. (a) Exponential decay
 (b) Time after purchase, t , measured in years
 (c) £32,000
 (d) £13,101.24
10. (a) £952.80
 (b) Assuming the employee earns the same amount for 52 weeks:

$$952.80 \times 52 = \text{£}49,545.60$$

11. (a) The branch probabilities are:

	First branches	Second branches
Bus	0.55	0.12 late, 0.88 not late
Not bus	0.45	0.04 late, 0.96 not late

- (b) 0.066
 (c) 0.084

12. (a) The completed regions are:

Region	Number of students
L only	27
C only	25
G only	15
$L \cap C$ only	24
$L \cap G$ only	19
$C \cap G$ only	13
$L \cap C \cap G$	8
None	11

(b) i. $\frac{27}{150} = \frac{9}{50}$
 ii. $\frac{56}{150} = \frac{28}{75}$
 iii. $\frac{80}{150} = \frac{8}{15}$
 iv. $\frac{139}{150}$

13. (a) Linear

(b) Time, t , measured in minutes

(c) Depth, d , measured in centimetres

(d) 2.5 centimetres per minute

(e) Approximately 93 centimetres

14. (a) Quadratic

(b) £59

15. For example, assume that:

- each person has one shower per day;
- each shower lasts 8 minutes;
- the shower uses 10 litres of water per minute.

$$5000 \times 7 \times 8 \times 10 = 2,800,000$$

Approximately 2.8 million litres.

16. (a) National Insurance is calculated using gross pay before deductions such as pension contributions.

(b) £270.18

17. (a) Time after launch, m , measured in months

(b) Number of subscribers, S

(c) Exponential growth

(d) 850 subscribers

(e) Approximately 2357 subscribers

18. (a) £50,400
(b) £9,181.80
(c) £38,127.60
19. (a) Time after the stall opens, w , measured in weeks
(b) Weekly profit, P , measured in pounds
(c) £550
20. (a) Linear
(b) 6 centimetres per minute
(c) The graph would have a steeper positive gradient and would still pass through the origin.
(d) The graph would have the same gradient but would cross the vertical axis at 10 centimetres.